

NAME: _____

PERIOD: _____

Two Analysts, One Planned Test

AP Statistics · Unit 3 · Topic 3.12

[STAR] THE PROBLEM

Suppose an online retailer has 50,000 active accounts. Before collecting data, its team asks, “Does a progress meter change the proportion of active accounts completing checkout?” It randomly selects 360 accounts and randomly assigns 180 to each screen. The table shows the outcomes. Let p_M and p_S be the corresponding population completion proportions.

Nia: “Use $H_0 : p_M - p_S = 0$ versus $H_a : p_M - p_S \neq 0$ because either direction is a change.”

Omar: “Because $\hat{p}_M$ is larger, use $H_a : p_M - p_S > 0$.”

Checkout outcomes

Screen	n	Complete	Other
Meter	180	126	54
Standard	180	108	72

FIRST TAKE — one sentence before you work the parts

When the team asks whether checkout changes, should a decrease count too? Use the original question to explain.

[BOOK] SET THE QUESTION, PARAMETERS, AND CONDITIONS (keep on the page)

Define population proportions before writing hypotheses. For no difference, $H_0 : p_1 - p_2 = 0$. Choose $H_a : p_1 - p_2 > 0$, < 0 , or $\neq 0$ from the research question set **before** viewing results; do not choose a tail because of the observed sign. Use a two-sample z -test for a difference of proportions with independent random samples or randomized groups. Check each 10 percent condition when sampling without replacement. Under H_0 , pool: $\hat{p}_c = (x_1 + x_2)/(n_1 + n_2)$; all four pooled expected counts must be at least 10.

Work (a)–(c).

DOK 1 (a) Define p_M and p_S in context. Calculate $\hat{p}_M$, $\hat{p}_S$, and the observed difference $\hat{p}_M - \hat{p}_S$.

DOK 2 (b) Name the test procedure. Calculate the pooled proportion and verify randomization, the 10 percent condition, and all four pooled expected counts.

DOK 3 (c) In 3–4 sentences, choose and state the hypotheses. Use the original word change and when the direction was chosen. Explain how Omar’s choice after seeing the result changes the question and could overstate the evidence.

Frame: The alternative is _____ because the original question asks _____.

Frame: Choosing the direction after seeing _____ could mislead readers by _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored **E / P / I**.